

Auditr Two-Person Presentation Script

Prepared for Abdul Hafeedh and Maryam Hafna using the current Auditr codebase.

Suggested duration: 5 to 7 minutes total.

Speaker 1: Abdul Hafeedh

Speaker 2: Maryam Hafna

Opening structure

- Abdul opens with the problem, objective, and architecture.
- Maryam explains the workflow, results, and practical value.
- Both of you stay ready for code, model, and process questions.

Full script

Part 1 - Abdul Hafeedh: Introduction and system overview

Good morning. Our project is **Auditr: AI-Powered Audit Intelligence for Suspicious Transaction Detection in Accounting Ledgers**.

The problem we focused on is that real accounting ledgers are often messy. They may contain invalid dates, invalid amounts, duplicate invoice patterns, inconsistent column names, and incomplete records. Many fraud-detection demos assume clean data, but real auditors do not get perfect files.

So our goal was not just to train a machine learning model. We wanted to build a practical audit workspace that can accept messy CSV ledgers, clean and validate them, score suspicious transactions, and then help an auditor understand why those transactions were flagged.

The project is structured into a frontend and a backend. The frontend is a Streamlit workspace with pages such as Home, Projects, Dashboard, Transactions, and Explainability. The backend handles authentication, project storage, CSV parsing, feature engineering, anomaly scoring, XGBoost inference, explainability, and export logic.

The workflow starts in the Projects page. The auditor creates an engagement and uploads a ledger CSV. Before any scoring happens, the system profiles the schema, maps column aliases, and quarantines invalid rows instead of letting broken data corrupt the analysis.

After that, the backend engineers historical and control-based features such as duplicate invoice indicators, vendor amount deviation, transaction bursts, invoice timing gaps, and approval-threshold patterns. These features are passed into the deployed XGBoost model, while a separate anomaly layer adds prioritization context.

I will now hand over to Maryam to explain the workflow outcome, results, and why this is useful from an audit perspective.

Part 2 - Maryam Hafna: Workflow, results, and conclusion

Once the analysis is complete, Auditr organizes the output into an auditor-friendly review flow instead of a raw machine-learning dashboard.

The Dashboard shows how many transactions were reviewed, how many were flagged, the review rate, the main suspicious reasons, the departments involved, vendor exposure, and the manual review queue.

The Transactions page acts as a ledger explorer. It allows filtering by review status, department, payment method, and risk threshold, and it also supports reviewer feedback such as cleared, needs review, or escalated.

The Explainability page focuses on one flagged case at a time. It provides the main reason, supporting signals, case facts, vendor-level evidence, and suggested next steps for the auditor. This is important because a flagged case should not be treated as confirmed fraud. It is only a prioritization signal for manual review.

In our current chronological holdout evaluation, the deployed XGBoost model achieved a test ROC-AUC of **0.989** and a test F1 score of **0.937**. In the high-risk demo engagement, the review rate was about **16.0%**, which kept the queue focused enough for practical manual investigation.

We also added anomaly-assisted prioritization. This helps highlight transactions that look unusual even if they do not perfectly match previously observed fraud patterns.

In conclusion, Auditr shows that AI-based audit support becomes much more useful when the model is integrated into a full workflow that includes data validation, explainability, project persistence, and reviewer-focused investigation.

Thank you. We would be happy to answer any questions.

Short fallback version if time is reduced

- Problem: messy ledger CSVs make traditional fraud review slow and unreliable.
- Solution: a project-based audit workspace that validates the file, quarantines bad rows, scores risk, and explains flagged transactions.
- Model result: XGBoost achieved ROC-AUC 0.989 and F1 0.937 on a chronological holdout test.
- Practical value: the auditor sees a focused review queue rather than raw probabilities only.